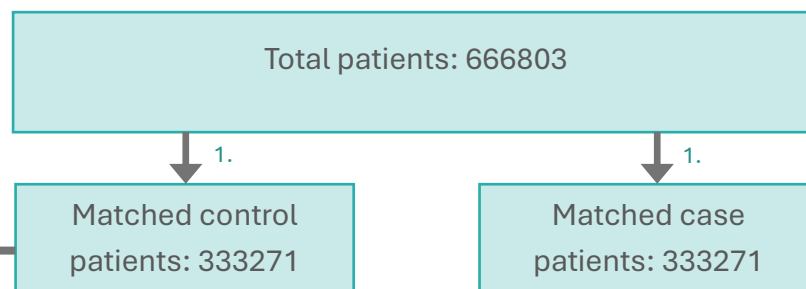


Cancer patients with pre-existing anxiety and/or depression

* : Number of patients dropped
NPD = No patients dropped
A/D = Anxiety and/or depression
OP = Outpatient appointment
S+S = Symptoms and treatment specialists

Step 1: Patients



3. Remove patients without a non-cancer matched control (*261)

Step 2: Cancer files

Control patients: 333271

5.

Control patients: 274053

2. Remove patients without a cancer matched case and those where date of death and/or reg end date is before case diagnosis date (*59218)

Cancer records: 353006

2a.

Cancer records: 352735

2b.

Cancer records/patients: 333271

3.

Cancer records/patients: 307580

4.

Cancer records/patients: 275087

Cancer patients: 333271

5. Ensure one cancer record per patient (NPD)

- Remove cancer records without a cancer code
- Multiple same day diagnoses: Select first cancer record, then highest stage cancer. Select random from remaining

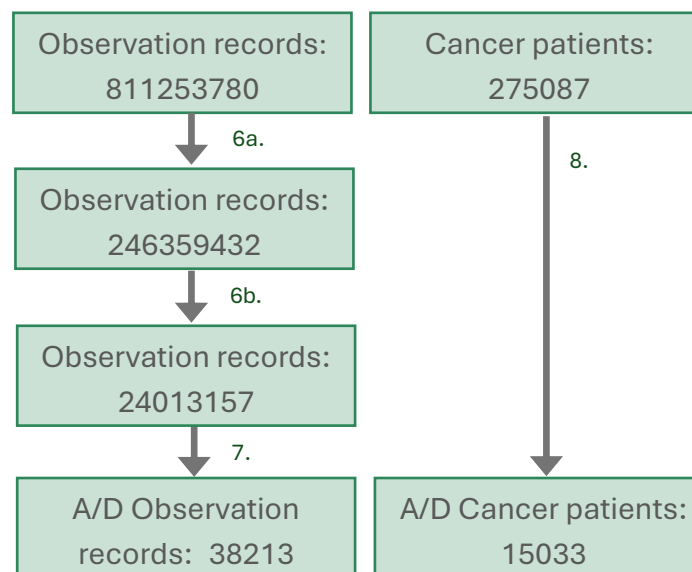
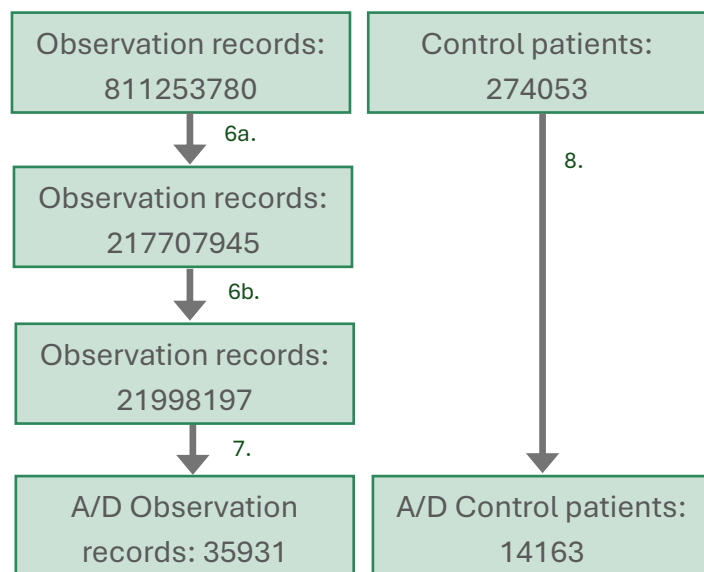
6. Remove patients with screening diagnosis route and aged under 40 at diagnosis (*25691)

7. Remove patients where date of death and/or reg end date is before diagnosis date. Reg date more than 3 years before diagnosis date (*32493)

Total cancer patients: **275087**
Total control patients: **274053**

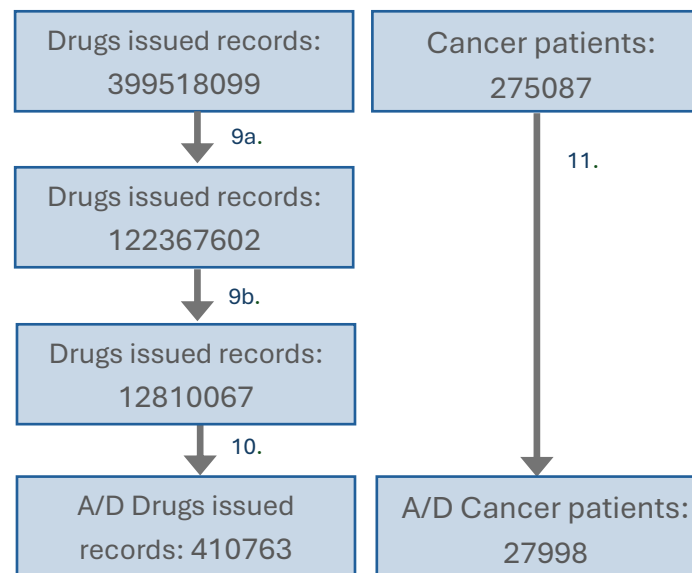
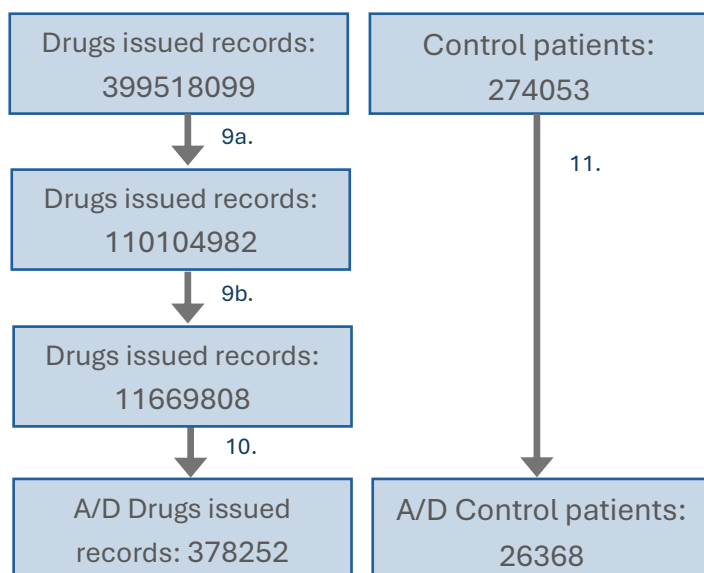
Next step: Anxiety and/or depression patients from observation records – Step 3a
Anxiety and/or depression patients from drug issues records – Step 3b

Step 3a: A/D from observation



6. Identify observations relating to pre-existing co-morbidities for patients in Step 2 (NPD)
 - a. Merge with cancer/control patients and remove observations after cancer diagnosis date
 - b. Retain all observations that are 1-2 years before cancer diagnosis date
7. Retain all observations that are related to anxiety or depression from code lists (NPD)
8. Retain patients with A/D observation records (case:*260054, con:*259890)

Step 3b: A/D from drugs issued



9. Identify drugs issued relating to pre-existing co-morbidities for patients in Step 2 (NPD)
 - a. Merge with cancer/control patients and remove issues after cancer diagnosis date
 - b. Retain all drugs issued 1-2 years before cancer diagnosis date
10. Retain all anxiety or depression drugs issued from drug code lists (NPD)
11. Retain patients who have at least 4 A/D drugs issued (case: *247089, con: *247685)

Total **cancer** patients with A/D observations: **15033**
 with A/D drugs issued: **27998** } Duplicates: 7273

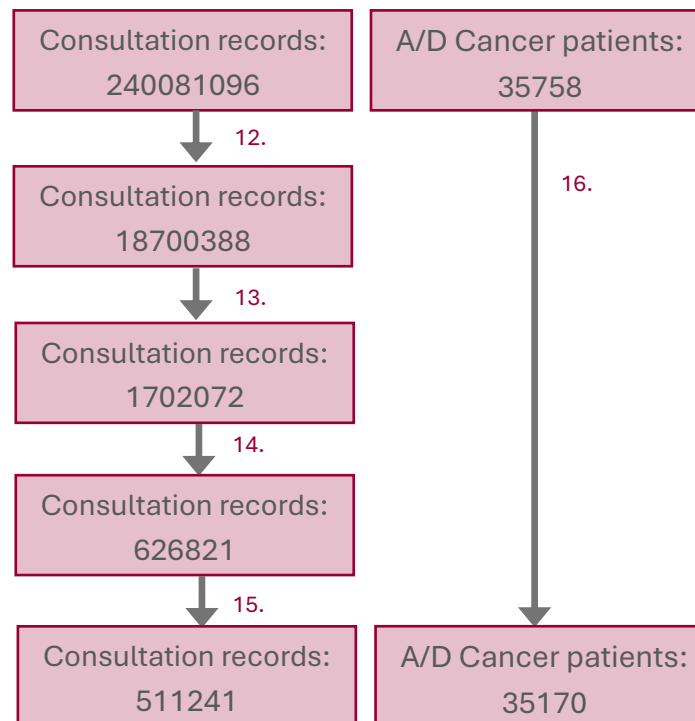
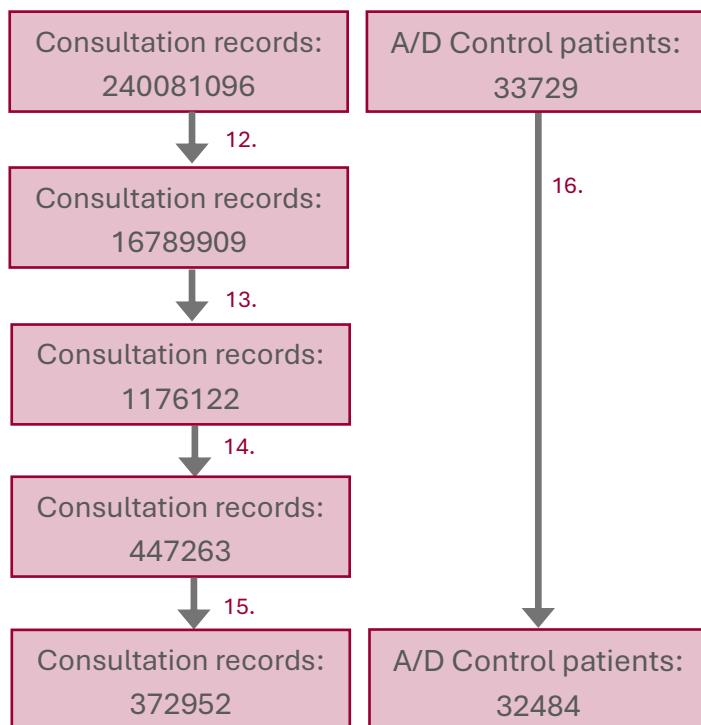
Total **control** patients with A/D observations: **14163**
 with A/D drugs issued: **26368** } Duplicates: 6802

Total A/D **cancer** patients: **35758**

Total A/D **control** patients: **33729**

Next step: Consultations

Step 4: Consultation files

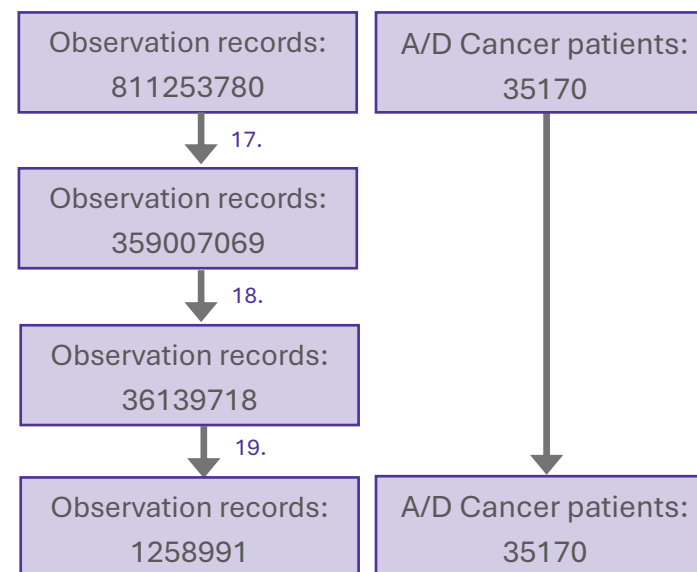
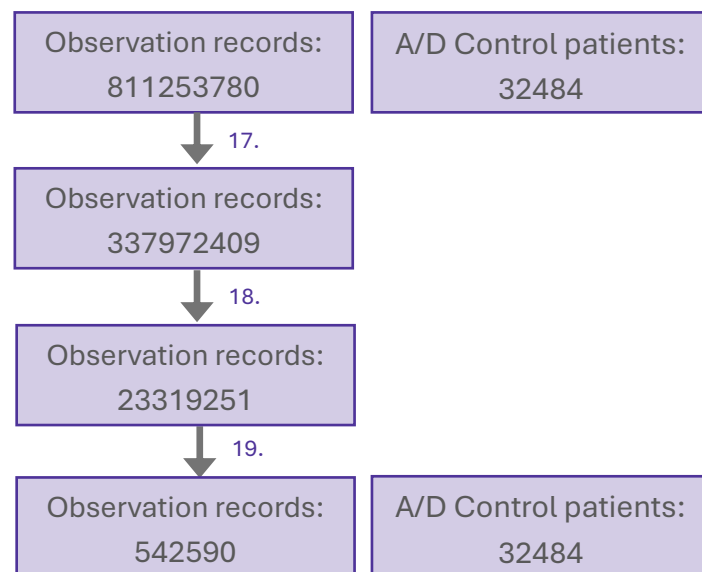


12. Merge A/D cancer/control patients with consultation files (NPD)
13. Retain all consultations within one year before cancer diagnosis (NPD)
14. Retain all consultations that are classed as face-to-face with a GP based on code list (NPD)
15. Remove multiple entries on the same day, retaining the first appointment recorded (NPD)
16. Retain only cancer patients who have at least one GP consultation (case: *588, con: *1245)

Total A/D **cancer** patients with at least one GP consultation: **35170**
Total A/D **control** patients with at least one GP consultation: **32484**

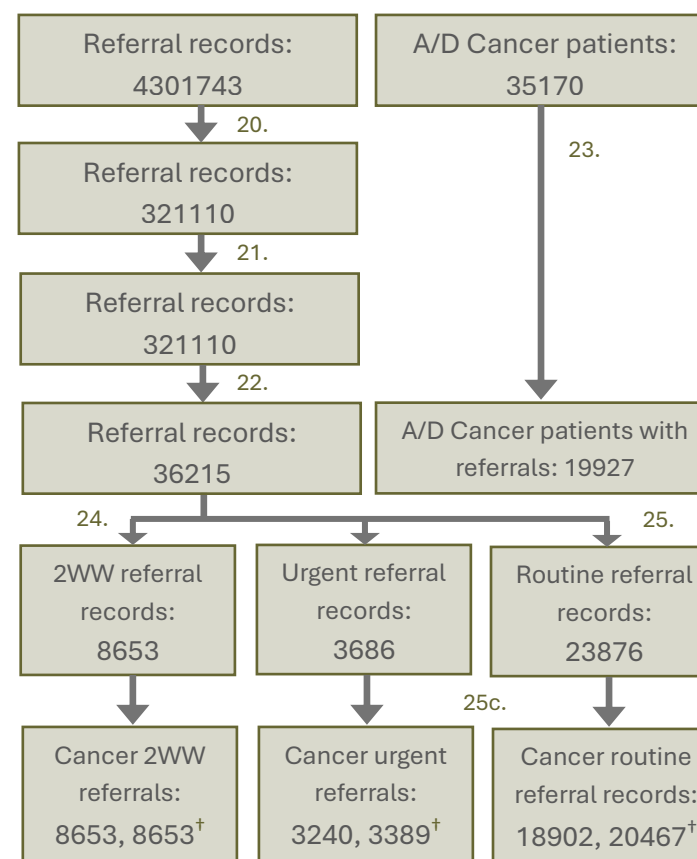
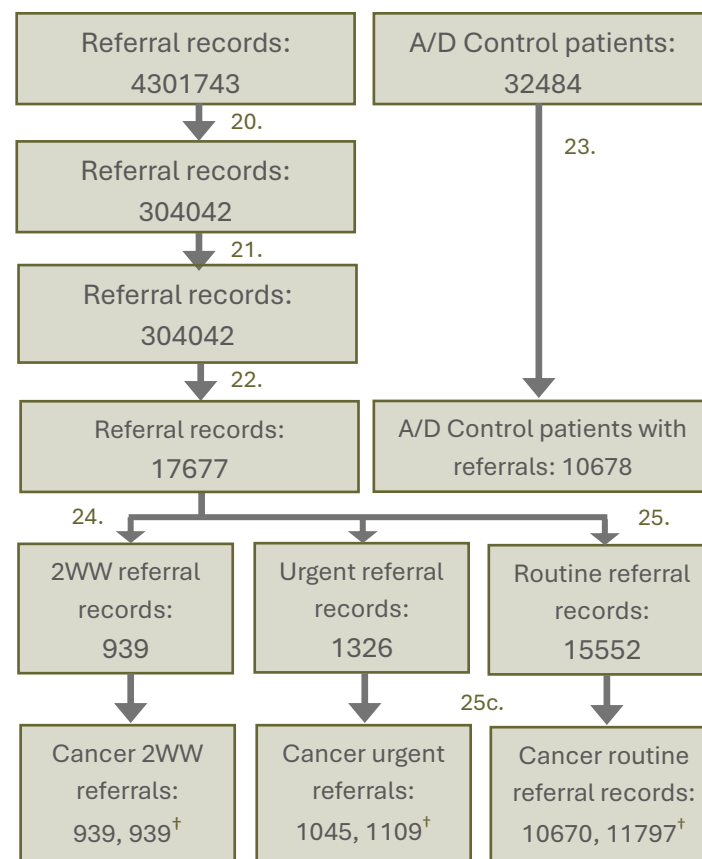
Next step: Identifying cancer referrals from cancer symptoms recorded as observations

Step 5: Symptoms from observations



17. Merge cancer/control patients with observation files (NPD)
18. Retain all observations that are within 1 year before cancer diagnosis date (NPD)
19. Retain all observations which contain either a symptom from cancer symptom code list or an observation identified as a referral (NPD)

Step 6: CPRD cancer referrals from symptoms in observation files



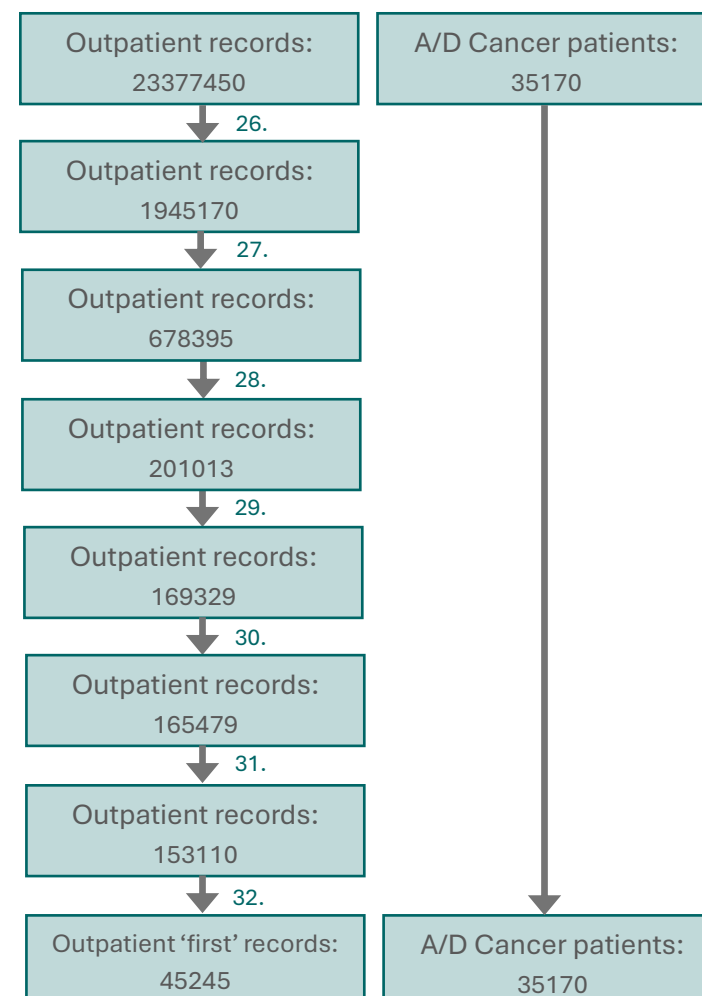
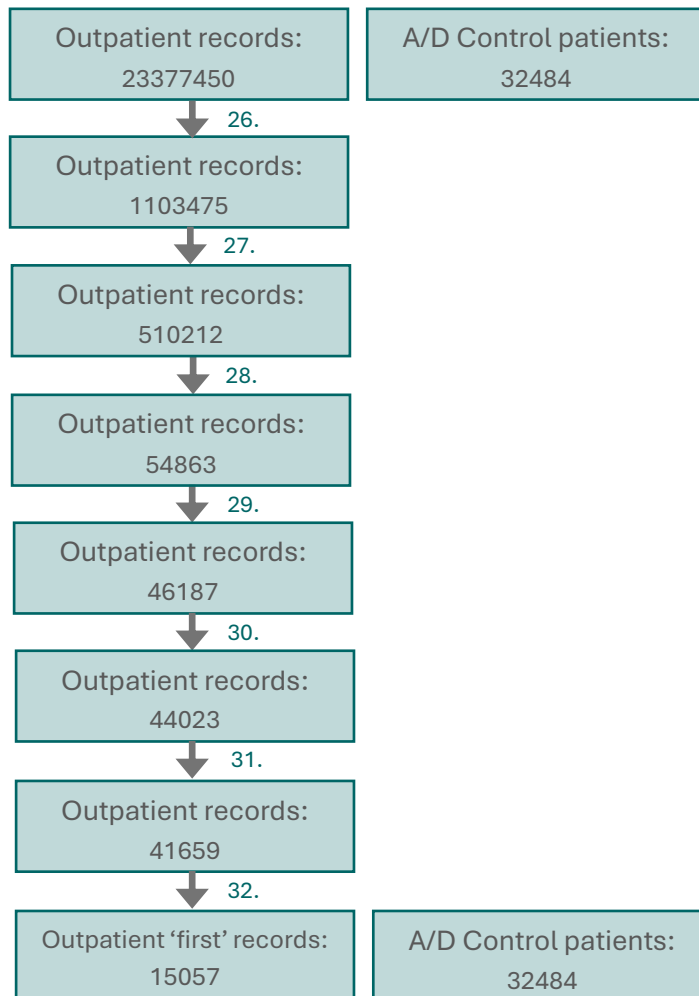
20. Merge cancer patients with CPRD referral file (NPD)
21. Merge cancer symptom observation records with CPRD referral file – ensures referrals are within 1 year before cancer diagnosis date (NPD)
22. Remove repeated cancer symptom observations (NPD)
23. Retain patients with at least one referral (case: *15243, con: *21806)
24. Referrals with 2WW priority are immediately taken to be cancer related (NPD)
25. For referrals with non-2WW priority (NPD):
 - a. Define symptom window around referral date[†]
 - b. Flag cancer symptom observations and GP consultations within window
 - c. Retain only referrals that have both an observation and a GP consultation within the specified window

Total CPRD cancer related 2WW referrals for A/D **cancer** patients: **8653, 8653[†]**
 Total CPRD cancer related urgent referrals for A/D **cancer** patients: **3240, 3389[†]**
 Total CPRD cancer related routine referrals for A/D **cancer** patients: **18902, 20467[†]**

Next step: Identifying cancer HES referrals and appointments from cancer symptoms recorded as observations and treatment specialists

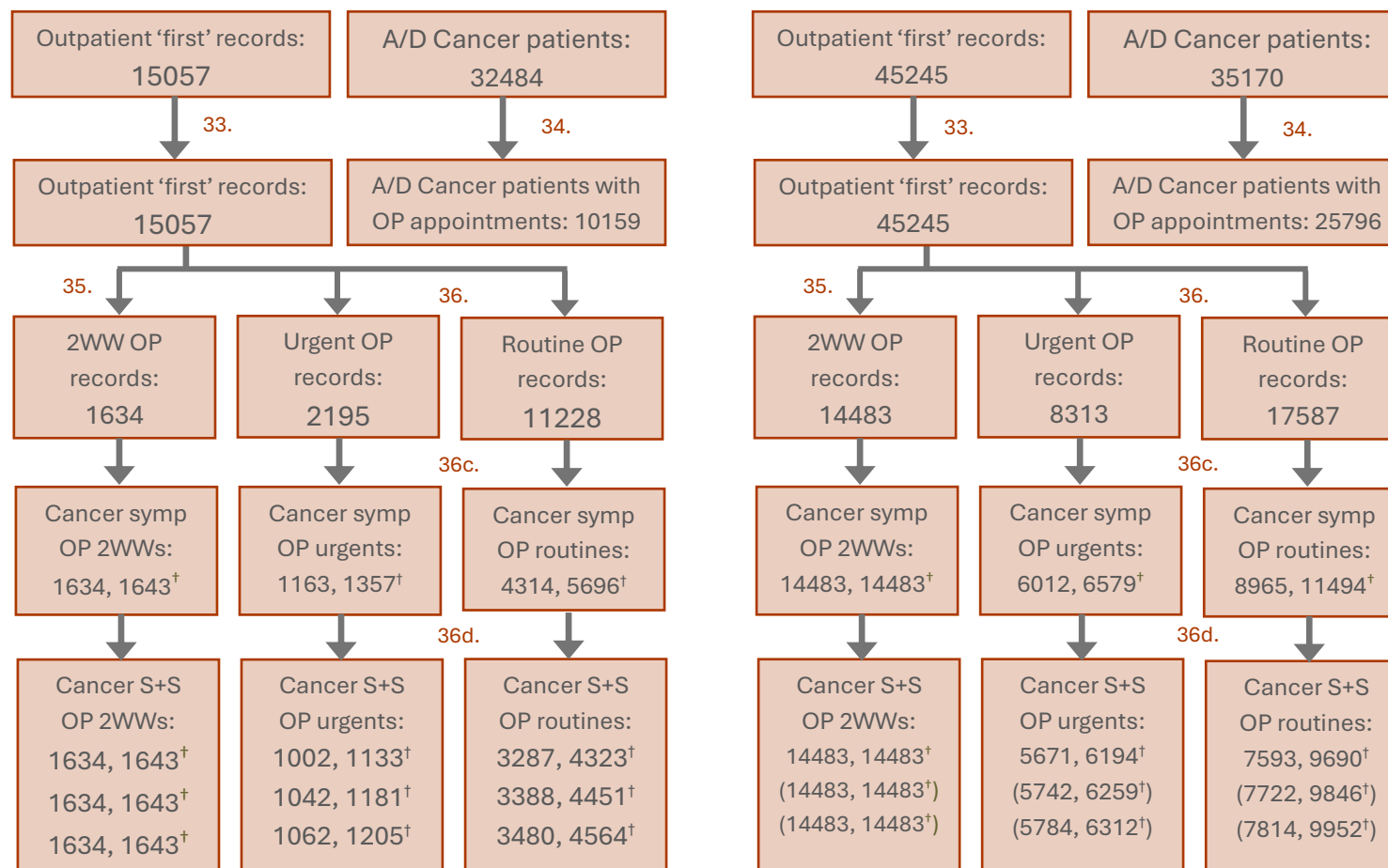
Total CPRD cancer related 2WW referrals for A/D **control** patients: **939, 939[†]**
 Total CPRD cancer related urgent referrals for A/D **control** patients: **1045, 1109[†]**
 Total CPRD cancer related routine referrals for A/D **control** patients: **10670, 11797[†]**

Step 7: HES Outpatient appointments



26. Merge cancer patients with HES outpatient appointment files (NPD)
27. Retain appointments that came from a GP referral (NPD)
28. Retain appointments that are requested from a GP within 1 year before cancer diagnosis date (NPD)
29. Remove appointments that were either cancelled, or unknown attendance (NPD)
30. Remove any 'follow-up' appointments which are not preceded by a 'first' appointment by date (NPD)
31. Remove any 'follow-up' appointments where requested date is before requested date of the corresponding 'first' appointment (NPD)
32. Remove any 'follow-up' appointments (recording total number corresponding to each 'first' appointment) (NPD)

Step 8: Cancer referrals from HES outpatient appointments



† window length: -7days, -14 days

33. Merge cancer symptom observation records with outpatient 'first' records (NPD)
34. Retain patients with at least one OP appointment (case: *9374, con: *22325)
35. Referral appointments with 2WW priority are immediately taken to be cancer related (NPD)
36. For appointments with a non-2WW priority:
 - a. Define symptom window around the appointment requested date (NPD)
 - b. Flag cancer symptom observations and GP consultation within window (NPD)
 - c. Retain appointments that have both an observation and a GP consultation within the specified window (NPD)
 - d. Retain appointments with flagged cancer symptoms that correspond with appointment treatment specialist (NPD)

() : Different sensitivity levels of cancer symptoms and corresponding treatment specialists

Total HES OP cancer related 2WW appointments for A/D **cancer** patients: **14483, 14483[†]**
 Total HES OP cancer related urgent appointments for A/D **cancer** patients: **5671, 6194[†]**
 Total HES OP cancer related routine appointments for A/D **cancer** patients: **7593, 9690[†]**

Total HES OP cancer related 2WW appointments for A/D **control** patients: **1634, 1634[†]**
 Total HES OP cancer related urgent appointments for A/D **control** patients: **1002, 1133[†]**
 Total HES OP cancer related routine appointments for A/D **control** patients: **3287, 4323[†]**